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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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Abstract

The AI-Powered Predictive Software Engineering Platform is a web-based application developed to support software development teams by providing intelligent predictions and analysis of software projects. Traditional software development often depends on manual estimation and analysis, which can make it difficult to identify project risks, defects, development effort, and potential delays at an early stage. This project aims to overcome these limitations by using Artificial Intelligence and Machine Learning techniques to analyze software project data and provide predictive insights.

The system allows users to manage software project information, analyze project metrics, monitor development progress, and obtain predictions related to software defects, project risks, development effort, and project performance. Based on historical project data, software metrics, and development information, the system can identify potential issues and provide useful predictions to support better decision-making. The predictions can help developers and project managers take appropriate actions at an early stage.

The application can be developed using React.js for designing an interactive and user-friendly frontend, FastAPI with Python for developing the backend and REST APIs, and PostgreSQL for storing project details, software metrics, prediction results, and related information. Machine learning techniques can be used to analyze historical software engineering data and generate predictions. The frontend communicates with the FastAPI backend to retrieve, manage, and display project information and prediction results.

The main objective of the AI-Powered Predictive Software Engineering Platform is to make software development more predictable, efficient, and reliable. By using AI-based predictions for defects, risks, effort, and project

performance, the system provides an intelligent approach to supporting software engineering activities and improving project planning and decision-making.

Signature of the Guide HOD